

Hospital Patient
Registration SystemLibrary Book
Management System

Reserved

Smart Hospital Patient Registration System

A multi-specialty hospital wants to digitize its outpatient registration system.

Every patient arriving at the hospital receives a registration number. The hospital management wants a Java application that maintains patient records efficiently while preventing duplicate registrations.

Each patient record consists of

☐ Patient ID

☐ Patient Name

☐ Mobile Number

☐ Department

☐ Doctor Name

The system should utilize Java Collection Framework and String manipulation techniques for storing, validating, searching, and displaying patient information.

Functional Requirements

Develop a Java program using ArrayList and HashMap to perform the following operations.

Menu

1. Register Patient

2. Search Patient

3. Display All Patients



Sample Input

1

P101

Rahul Kumar

9876543210

Cardiology

Dr.Arun

1

P102

Priya

9876543211

Ortho

Dr.Meena

3



Q Search

ENG
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JAVA PROGRAMMING

Akileshwaran A
192511354RD

5

Sample Output

Patient Registered Successfully.

Patient Registered Successfully.

ID Name Department

P101 Rahul Kumar CARDIOLOGY

P102 Priya ORTHO

Department Count

CARDIOLOGY : 1

ORTHO : 1

Patient Details

ID : P101

Name : Rahul Kumar



Q Search



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Your Code

```
1 import java.util.*;
2
3 public class Main {
4     static ArrayList<String> patientList = new ArrayList<>();
5     static HashMap<String, String[]> patientMap = new HashMap<>();
6     static HashMap<String, Integer> deptCount = new HashMap<>();
7     static final int MAX_PATIENTS = 100;
8
9     public static void main(String[] args) {
10         Scanner sc = new Scanner(System.in);
11         int choice;
12
13         do {
14             System.out.println("1. Register Patient");
15             System.out.println("2. Search Patient");
16             System.out.println("3. Display All Patients");
17             System.out.println("4. Count Department-wise Patients");
18             System.out.println("5. Exit");
19
20             choice = Integer.parseInt(sc.nextLine().trim());
21
```

Input

P101
Rahul Kumar
9876543210
Cardiology
Dr. Arun

Output



Search



ENG
IN



```
20     choice = Integer.parseInt(sc.nextLine().trim());
21
22     switch (choice) {
23         case 1:
24             registerPatient(sc);
25             break;
26         case 2:
27             searchPatient(sc);
28             break;
29         case 3:
30             displayAllPatients();
31             break;
32         case 4:
33             departmentWiseCount();
34             break;
35         case 5:
36             System.out.println("Exiting...");
37             break;
38         default:
39             System.out.println("Invalid Choice");
40     }
41     } while (choice != 5);
42
```



Search

ENG
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```
30         System.out.println("Invalid Choice");
31     }
32     while (choice != 0) {
33     }
34     sc.close();
35 }
36
37 static void registerPatient(Scanner sc) {
38     if (patientMap.size() >= MAX_PATIENTS) {
39         System.out.println("Maximum Patient Limit Reached.");
40         return;
41     }
42
43     System.out.print("\n");
44     String id = sc.nextLine().trim();
45
46     if (patientMap.containsKey(id)) {
47         System.out.println("Duplicate Patient ID, Registration
48         not possible.");
49     }
50
51     String name = sc.nextLine().trim();
```



```

62         System.out.println("Invalid Name. Only alphabets allowed");
63         return;
64
65     name = toProperCase(name);
66
67     String mobile = sc.nextLine().trim();
68     if (!mobile.matches("[0-9]{10}")) {
69         System.out.println("Invalid Mobile Number. Must be 10 digits");
70         return;
71     }
72
73     String department = sc.nextLine().trim().toUpperCase();
74
75     String doctor = sc.nextLine().trim();
76
77     String[] patientData = {id, name, mobile, department, doctor};
78     patientMap.put(id, patientData);
79     patientList.add(id);
80
81     deptCount.put(department, deptCount.getOrDefault(department, 0) + 1);
82
83     System.out.println("Patient Registered Successfully.");
84

```



Search

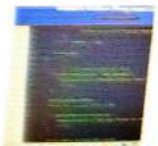



```

86  static void searchPatient(Scanner sc) {
87      System.out.print("");
88      String id = sc.nextLine().trim();
89
90      if (patientMap.containsKey(id)) {
91          String[] p = patientMap.get(id);
92          System.out.println("Patient Details");
93          System.out.println("ID : " + p[0]);
94          System.out.println("Name : " + p[1]);
95          System.out.println("Mobile : " + p[2]);
96          System.out.println("Department : " + p[3]);
97          System.out.println("Doctor : " + p[4]);
98      } else {
99          System.out.println("Patient Not Found");
100     }
101 }
102
103 static void displayAllPatients() {
104     System.out.println("-----");
105     System.out.println("ID\tName\tMobile\tDepartment\tDoctor");
106     System.out.println("-----");
107
108     for (String id : patientList) {

```

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via Mo



Search




```
109         String[] p = patientMap.get(id);
110         System.out.println(p[0] + "\t" + p[1] + "\t" + p[2] + "
111     }
112 }
113
114 static void departmentWiseCount() {
115     System.out.println("Department Count");
116     for (Map.Entry<String, Integer> entry : deptCount.entrySet()
117         System.out.println(entry.getKey() + " : " + entry.getVa
118     }
119 }
120
121 static String toProperCase(String str) {
122     String[] words = str.trim().split("\\s+");
123     StringBuilder result = new StringBuilder();
124
125     for (String word : words) {
126         if (word.length() > 0) {
127             result.append(Character.toUpperCase(word.charAt(0))
128                 .append(word.substring(1).toLowerCase())
129                 .append(" ");
130         }
```


QUESTIONS

Q1Smart Hospital Patient
Registration System
20 marks**Q2**Online Library Book
Management System
20 marks**Q2 Online Library Book Management System**

20 marks

Scenario**Online Library Book Management System**

A university library is planning to automate its book management process. The librarian wants a Java application that stores book details, prevents duplicate book entries, allows students to search for books quickly, and generates category-wise reports. The application should use Java String handling and Java Collection Framework to efficiently manage the library records.

Requirements

Each book contains the following details:

- ☐ Book ID
- ☐ Book Title
- ☐ Author Name
- ☐ Category
- ☐ Publisher

Use:

- ☐ ArrayList to store all book objects.
- ☐ HashMap to store Book ID as the key for quick searching.
- ☐ HashSet to maintain unique author names.

2/2 answered



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 **Akileshwaran A**
192511354RD

Sample Input

1

B101

Java Programming

Herbert Schildt

Programming

McGraw Hill

1

B102

Data Structures

Mark Allen

Computer Science

Pearson

3



Search



ENG
IN



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Sample Output

Book Added Successfully.

Book Added Successfully.

Book ID Title Author-----
B101 Java Programming Herbert Schildt

B102 Data Structures Mark Allen

Unique Authors

Herbert Schildt

Mark Allen

Category-wise Count

PROGRAMMING : 1

COMPUTER SCIENCE : 1

Book Details



Search

ENG
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Your Code

Input

1
8101
Herbs
Progr

Output

```
1 #include <iostream>
2 using namespace std;
3
4 struct Book {
5     string title;
6     string author;
7     string publisher;
8 }
9
10 int main() {
11     Book b1, b2, b3;
12     b1.title = "C++";
13     b1.author = "Herbs";
14     b1.publisher = "Progr";
15     b2.title = "C++";
16     b2.author = "Herbs";
17     b2.publisher = "Progr";
18     b3.title = "C++";
19     b3.author = "Herbs";
20     b3.publisher = "Progr";
21     return 0;
22 }
```



21

```
22  
23  
24 static String trimSafe(String s) {  
25     if (s == null) return null;  
26     return s.trim();  
27 }  
28  
29  
30  
31 static String toTitleCase(String s) {  
32     s = trimSafe(s);  
33     if (s.length() == 0) return s;  
34  
35     s = s.toLowerCase();  
36     StringBuilder out = new StringBuilder();  
37     boolean newWord = true;  
38  
39     for (int i = 0; i < s.length(); i++) {  
40         char ch = s.charAt(i);  
41         if (ch == ' ') {  
42             out.append(" ");  
43             newWord = true;  
44         }  
45         else {  
46             if (newWord) {  
47                 out.append(Character.toUpperCase(ch));  
48                 newWord = false;  
49             }  
50             else {  
51                 out.append(ch);  
52             }  
53         }  
54     }  
55     return out.toString();  
56 }
```



Search




```
49 return out.toString().replaceAll("\\t", " ");
```

```
50  
51 public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    ArrayList<Book> books = new ArrayList<>();
```

```
    HashMap<String, Book> bookById = new HashMap<>();
```

```
    HashSet<String> authors = new HashSet<>();
```

```
    int maxBooks = 1000;
```

```
    while (true) {
```

```
        if (!sc.hasNextLine()) break;
```

```
        String line = sc.nextLine().trim();
```

```
        if (line.length() == 0) continue;
```

```
        int choice
```

```
        choice = Integer.parseInt(line);
```

```
        switch (choice) {
```



Search




```
if (choice == 1) break;
```

```
if (choice == 2) {
```

```
    if (books.size() >= maxBooks) continue;
```

```
    String id = trimSafe(sc.nextLine());
```

```
    String title = trimSafe(sc.nextLine());
```

```
    String author = trimSafe(sc.nextLine());
```

```
    String category = trimSafe(sc.nextLine());
```

```
    String publisher = trimSafe(sc.nextLine());
```

```
    boolean ok = true;
```

```
    if (id.length() == 0) ok = false;
```

```
    if (title.length() == 0) ok = false;
```

```
    if (author.length() == 0) ok = false;
```

```
    if (category.length() == 0) ok = false;
```

```
    if (publisher.length() == 0) ok = false;
```

```
    if (bookById.containsKey(id)) ok = false;
```




```

106
107     } else if (choice == 2) {
108         String id = trimSafe(sc.nextLine());
109         Book b = bookById.get(id);
110
111         if (b == null) {
112             System.out.println("Book Not Available");
113         } else {
114             System.out.println("Book ID : " + b.id);
115             System.out.println("Book Title : " + b.title);
116             System.out.println("Author : " + b.author);
117             System.out.println("Category : " + b.category);
118             System.out.println("Publisher : " + b.publisher);
119         }
120
121     } else if (choice == 3) {
122         System.out.println("-----");
123         System.out.println("ID TITLE AUTHOR CATEGORY");
124         System.out.println("-----");
125         for (int i = 0; i < books.size(); i++) {
126             Book b = books.get(i);
127             System.out.println(b.id + " " + b.title + " " +
128

```

12301



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phot
via M



Search



```

1271         System.out.println();
1272         for (int i = 0; i < books.size(); i++)
1273             Book b = books.get(i);
1274             System.out.println(b.id + " " + b.title + " " + b.author);
1275
1276         } else if (choice == 2)
1277             System.out.println("Unique Authors");
1278             for (String a : authors)
1279                 System.out.println(a);
1280
1281         } else if (choice == 3)
1282             HashMap<String, Integer> catCount = new HashMap<>();
1283
1284             for (int i = 0; i < books.size(); i++)
1285                 Book b = books.get(i);
1286                 String c = b.category;
1287
1288                 Integer old = catCount.get(c);
1289                 if (old == null)
1290                     catCount.put(c, 1);
1291                 else
1292                     catCount.put(c, old + 1);
1293
1294         }
1295     }
1296 }

```


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```

148 System.out.println("Category-wise Count");
149 for (String key : catCount.keySet()) {
150     System.out.println(key + " : " + catCount.get(key));
151 }
152
153 for (int i = 0; i < books.size(); i++) {
154     Book b = books.get(i);
155     System.out.println("Book Details");
156     System.out.println("Book ID : " + b.id);
157     System.out.println("Title : " + b.title);
158     System.out.println("Author : " + b.author);
159     System.out.println("Category : " + b.category);
160     System.out.println("Publisher : " + b.publisher);
161
162     String t = b.title.trim();
163     int len = b.title.length();
164
165     int words = 0;
166     for (int i = 0; i < len; i++) {
167         words++;
168         if (Character.isAlphabetic(t.charAt(i)) == false) words++;
169     }
170

```

12301



Q Search



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```

141 int len = b.title.length();
142
143 int words = 0;
144 while (t.length() > 0) {
145     words = 1;
146     for (int k = 0; k < t.length(); k++) {
147         if (t.charAt(k) == ' ') words++;
148     }
149 }
150
151 System.out.println("Length : " + len);
152 System.out.println("Words : " + words);
153 System.out.println("Category : " + b.category);
154
155
156
157
158
159
160
161
162

```



Search

